UNIVERSITI TEKNOLOGI MALAYSIA

SECP3843

Special Topic in Data Engineering

Tutorial MS Azure

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1. Project Overview

A company has recognized a gap in understanding its customer demographics, specifically the gender distribution within the customer base and how it might influence product purchases. With a significant amount of customer data stored in an on-premises SQL database, key stakeholders have requested a comprehensive KPI dashboard. This dashboard should provide insights into sales by gender and product category, showing total products sold, total sales revenue, and a clear gender split among customers. Additionally, they need the ability to filter this data by product category and gender, with a user-friendly interface for date-based queries.

The objective of this project is to address the challenge of building a scalable, automated end-to-end data pipeline using Azure services to ingest raw data from the on-premise database and transform it into data that can be visualized to determine actionable insights. This project is relevant as it simulates a real-world scenario where an enterprise requires to convert raw, unstructured data into actionable intelligence that can drive strategic decision-making.The scope of this project covers the entire data lifecycle from ingestion, storage, transformation, and visualization. It begins with the ingestion of raw SQL data from Microsoft's AdventureWorks sample databases and visualizing into a dashboard in Power BI as the final product.

2. Description on the solution

To address this request, we'll build a robust data pipeline that extracts the on-premises data, loads it into Azure, and performs the necessary transformations to make the data more query-friendly. The transformed data will then feed into a custom-built report that meets all the specified requirements. This pipeline will be scheduled to run automatically every day, ensuring that stakeholders always have access to up-to-date and accurate data.

3. Technology Use

The table below shows all technology used to make this pipeline work seamlessly.

Technology	Functionality
SQL Server (On-Prem)	Primary source containing AdventureWorks2025 lightweight dataset.
Azure Self-Hosted Integration Runtime (SHIR)	The bridge between on-premises and the cloud. Installed on the local machine to allow Azure Data

	Factory to securely bypass firewalls and pull data from the local SQL Server without exposing the database to the public internet.
Azure Data Factory (ADF)	Act as the orchestrator. Uses Copy Activities to ingest raw tables into the cloud. It also manages the Control Flow, triggering Databricks notebooks and Synapse loads in a specific sequence.
Azure Data Lake Storage (ADLS)	The centralized data repository and organized into a Medallion Architecture. Bronze for raw Parquet/CSV files directly from the source. Silver for cleaned data with correct data types and removed duplicates. Finally gold, flattened business-ready tables optimized for the dashboard.
Azure Key Vault	Act as security management by storing Secrets (database passwords and connection strings). This ensures no sensitive credentials are hardcoded in ADF or Databricks notebooks for security purposes.
Azure Databricks	Use notebooks to mount the Data Lake. Databricks is used to perform complex transformation and cleaning here with Python. This results is transformed and ready to go for an analytical purpose table.
Azure Synapse Analytics	Acts as the high-speed serving layer. The Gold data from the Data Lake is loaded here using the COPY command, allowing Power BI to query with low latency.
Power BI	Business Intelligence and Storytelling. Connects to Azure Synapse via Import Mode. It transforms the Gold table into an interactive story by making dashboards.

4. Data Pipeline Architecture

The workflow starts with Azure Data Factory (ADF), which serves as the orchestration engine for ingesting raw data into Azure Data Lake Storage Gen2 (ADLS Gen2). The storage solution takes a layered approach, with the bronze layer capturing data in its original format to maintain data provenance. The data is then processed using Azure Databricks, where PySpark is used to conduct data transformations such as data cleaning before being moved to the silver layer. This design decision is crucial, as Databricks delivers the distributed computing capacity required to handle massive datasets significantly more effectively than traditional SQL-based solutions.

The next stage of the procedure is to modify the data in the gold layer, where the data is used for analytical querying. This filtered data is then delivered to Azure Synapse Analytics which acts as a data warehouse and then to Power BI for final visualization. The argument for this design is the separation of storage and computing, which allows for independent scalability and considerable cost reductions. By using ADLS Gen2 as the primary repository and ADF for automation, the system minimizes manual intervention, lowers the risk of human mistake and reduces silos that enables users to extract meaningful, visual insights from complicated raw data. Figure 4.1 is the suggested solution based on the Azure data architecture, which was chosen for its smooth integration and scalability.

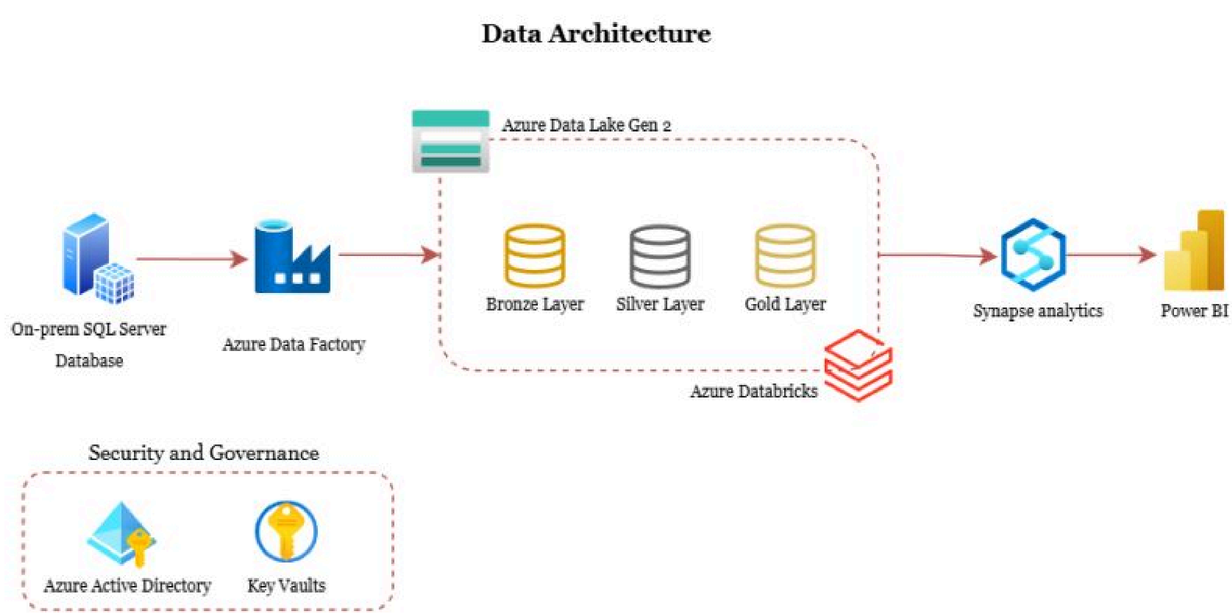

Figure 4.1: Azure Data Pipeline Architecture

5. MS Power BI Dashboard

The final reporting is presented as a two-page interactive dashboard that delivers a detailed overview of sales performance alongside customer demographic insights.

Dashboard 1: Sales Performance & Customer Demographics

This main dashboard addresses the core objective of analyzing the relationship between product sales and gender distribution. It features key performance indicators (KPIs) that provide a quick summary for stakeholders, including Total Sales ($708.69K), Total Products Sold (2,087), and the total number of products available (297).

A donut chart illustrates the gender distribution of customers based on titles such as Mr., Ms., Sr., and Sra., offering a clear view of the customer base. Sales performance is further analyzed through a vertical bar chart that ranks product categories by revenue, highlighting top-performing categories such as Touring Bikes and Road Bikes. In addition, a horizontal bar chart presents the quantity of products sold by category, enabling comparison between high-volume and high-revenue items.

To enhance usability, an interactive slicer for product categories allows users to dynamically filter all visuals on the dashboard.

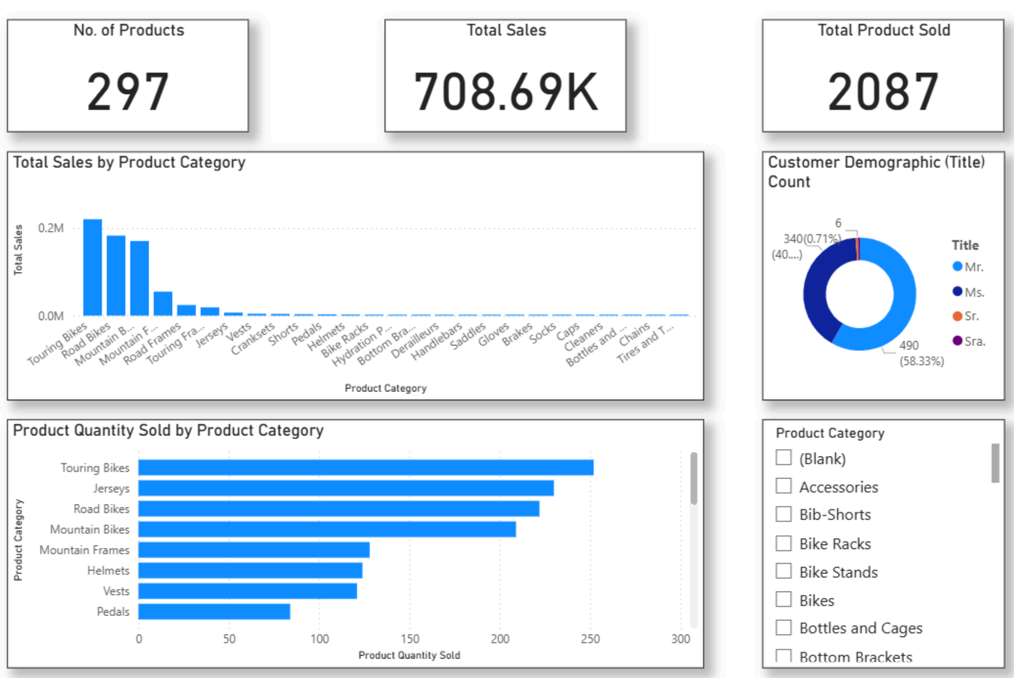

Figure 5.1: Sales Performance and Customer Demographics Dashboard

Dashboard 2: Geographic Sales Distribution

The second dashboard focuses on the geographical aspect of sales performance, helping to identify key markets. A donut chart shows the distribution of total sales by region, indicating that the United Kingdom contributes 60.11% ($425.98K) of total sales, while the United States accounts for 39.89% ($282.71K).

A more detailed breakdown is provided through a pie chart displaying sales by city, where London is shown to be the leading contributor with 21.69% ($153.75K) of total sales. Users can interact with this chart by tapping a city segment to filter and update all related visuals, enabling a more focused analysis at the region and city level. A geospatial map visual further enhances understanding by showing sales concentrations across North America and Europe.

Additionally, a scatter plot is included to analyze the relationship between total sales and product quantity sold by city, allowing stakeholders to identify trends and high-performing outliers.

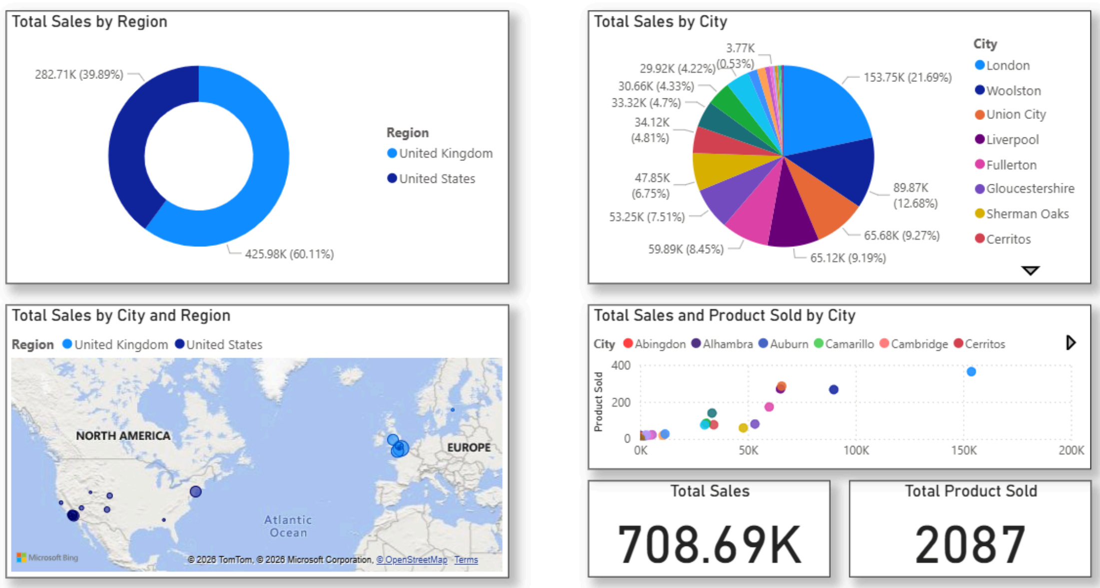

Figure 5.2: Geographical Aspect of Sales Performance Dashboard

6. Reflection

Iman Abadi bin Mohd Nizwan

After completing this project, my knowledge in data analytics and big data has changed from basic data manipulation to the building of automated, scalable systems. By building a tiered pipeline, I learned how a structured approach converts raw, individual and separated records into a single source of truth. The project demonstrated the effectiveness of cloud technologies such as Azure in separating storage and computing, assuring the system's flexibility and cost-effectiveness. Finally, I've realized that data engineering is important in creating a robust infrastructure that transforms raw data into a reliable insight for strategic decision-making.

Mohamed Alif Fathi bin Abdul Latif

This project helped me understand how a real data pipeline works from start to finish. I learned how to move data from an on-premises database into the cloud, clean and transform it, and finally present it in a dashboard. At first, the process felt complex, but breaking it into stages (bronze, silver, gold) made it easier to manage. I also realized the importance of automation and using the right tools together, such as Azure Data Factory, Databricks, and Power BI. This project improved my understanding of how data can be turned into meaningful insights for decision-making. Overall, it gave me practical experience and a clearer picture of real-world data engineering workflows.